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Absorbing boundary condition for the elastic wave equation

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ABSTRACT

Extant absorbing boundary conditions for the elastic wave equation are generally effective only for waves nearly normally incident upon the boundary. High reflectivity is exhibited for waves traveling obliquely to the boundary. In this paper, a new and efficient absorbing boundary condition for two-dimensional and three-dimensional finite-difference calculations of elastic wave propagation is presented. Compressional and shear components of the incident vector displacement fields are separated by calculating intermediary scalar potentials, allowing the use of Lindman's boundary condition for scalar fields, which is highly absorbing for waves incident at any angle. The elastic medium is assumed to be homogeneous in the region immediately adjacent to the boundary. The reflectivity matrix of the resulting absorbing boundary for elastic waves is calculated, including the effects of finite-difference truncation error. For effectively all angles of incidence, reflectivities are much smaller than those of the commonly employed paraxial absorbing boundaries, and the boundary condition is stable for any physical Poisson's ratio. The nearly complete absorption predicted by the reflectivity matrix calculations, even at near grazing incidence, is demonstrated in a finite-difference application.

INTRODUCTION

In finite-difference calculations of wave propagation phenomena, artificial reflections arise at the edges of the domain of computation. In order that these spurious reflections do not compromise the accuracy of results in interior regions, the domain of computation can be enlarged so that reflected signals cannot reach the region of interest within the given time period. Alternatively, "absorbing" boundary conditions may be employed. The first approach wastes computational resources, often prohibitively so for two-dimensional (2-D) and three-dimensional (3-D) computations, while extant absorbing boundary conditions generally reduce the amplitude of the reflected waves to an acceptable level only for waves nearly normally incident upon the boundary. Very high reflectivities may be exhibited for waves traveling obliquely to the boundary. For many applications this level of performance is not adequate, and the computational mesh must still be expanded to delay the arrival of the imperfectly absorbed signals.

Many formulations of absorbing boundary conditions for elastic finite-difference calculations have been proposed. Among these are viscous boundary conditions (Lysmer and Kuhlemeyer, 1969), paraxial boundary conditions (Clayton and Engquist, 1977; Engquist and Majda, 1979; Wagatha, 1983) and, recently, "multitransmitting" boundary conditions (Liao et al., 1984). The most widely used schemes for elastic waves appear to be those based upon paraxial approximations to the elastic wave equation. These boundary conditions perfectly absorb elastic waves normally incident on the boundary, but only partially absorb obliquely incident waves. Reflected wave amplitudes generally increase with increasing angle of incidence. Complete reflection occurs at grazing incidence. If acceptable performance is loosely defined to be the ratio of reflected wave to incident wave amplitude of a few percent or less, then the paraxial schemes generally become unacceptable for incident angles between about 30° and 45°, depending upon the Poisson's ratio of the elastic medium and incident wave type (compressional or shear) under consideration. In addition, there is empirical evidence of instability in the paraxial boundaries for Poisson's ratios larger than about 0.37 (Emerman and Stephen, 1983; Mahrer, 1986).

In this paper, an efficient absorbing boundary condition is described for multidimensional finite-difference calculations of elastic wave propagation. The boundary condition is based upon a generalization of the elegant scheme of Lindman (1975) for scalar waves. In principle, reflection coefficients may be reduced to 1 percent or less for angles of incidence up to 89°. In practice, finite-difference truncation error slightly degrades the performance of the boundary condition. The

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boundary condition absorbs evanescent as well as propagating waves and does not exhibit instability for large Poisson's ratios. The elastic medium is assumed to be homogeneous in the region immediately adjacent to the boundary. For later reference, a brief recapitulation of the 2-D scalar absorbing boundary condition of Lindman is given; details may be found in the original publication.

REVIEW OF LINDMAN'S ABSORBING BOUNDARY FOR SCALAR WAVES

Consider a plane scalar wave incident on the boundary $x = x_{\max}$ from the left, as shown in Figure 1a. The angle of incidence measured from the normal to the boundary is θ. The value of the scalar field is denoted by $\phi(x, y, t) = \Phi(x \cos\theta + y \sin\theta - ct)$, where y is the dimension transverse to x (along the boundary), c is the propagation speed of scalar waves, and t denotes time. A boundary condition which absorbs the incident scalar wave perfectly is

$$\frac{\partial\phi}{\partial t} + \frac{c}{\cos\theta}\frac{\partial\phi}{\partial x} = 0. \tag{1}$$

Most often, however, the incident field ϕ comprises an unknown spectrum of incident plane waves. Written as a sum of Fourier components,

$$\phi = \sum_{\omega}\sum_{\theta} \tilde{\phi}_{\omega\theta}(x)e^{i(k_y y - \omega t)}. \tag{2}$$

Here $k_y = k \sin\theta$, $k = \omega/c$, and $\tilde{\phi}_{\omega\theta}(x) = \tilde{\phi}_{\omega\theta}(0)\exp(ik_x x)$, where $k_x = k\cos\theta$. A generalization of equation (1) which contains no reference to θ, and therefore applies for an arbitrary incident field, is the desired result. For each individual Fourier component, a rational approximation is made for the term $(\cos\theta)^{-1}$ in equation (1),

$$\frac{1}{\cos\theta} \approx 1 + \sum_{m=1}^{m=M} \frac{\alpha_m \sin^2\theta}{1 - \beta_m \sin^2\theta}. \tag{3}$$

This approximation may be written as

$$\frac{1}{\cos\theta} \approx 1 + \sum_{m=1}^{m=M} \frac{\alpha_m (ck_y/\omega)^2}{1 - \beta_m (ck_y/\omega)^2}, \tag{4}$$

so that for each Fourier component $\tilde{\phi}_{\omega\theta}$ the absorbing boundary condition of equation (1) becomes

$$-i\omega\tilde{\phi}_{\omega\theta} + c\frac{\partial\tilde{\phi}_{\omega\theta}}{\partial x} = -\sum_{m=1}^{m=M}\tilde{h}_m, \tag{5}$$

where

$$\tilde{h}_m = \frac{\alpha_m c^2 k_y^2}{\omega^2 - \beta_m c^2 k_y^2}\left(c\frac{\partial\tilde{\phi}_{\omega\theta}}{\partial x}\right). \tag{6}$$

Clearing the denominator in equation (6) and associating $-i\omega \leftrightarrow \partial/\partial t$, $ik_y \leftrightarrow \partial/\partial y$ (inverting the Fourier transforms) yield the space-time domain absorbing boundary condition sought:

$$\frac{\partial\phi}{\partial t} + c\frac{\partial\phi}{\partial x} = -\sum_{m=1}^{m=M} h_m, \tag{7}$$

where

$$\frac{\partial^2 h_m}{\partial t^2} - \beta_m c^2\frac{\partial^2 h_m}{\partial y^2} = \alpha_m c^2\frac{\partial^2}{\partial y^2}\left(c\frac{\partial\phi}{\partial x}\right). \tag{8}$$

This boundary condition applies to each plane-wave component in the spectrum independently, and therefore, to ϕ itself. Thus, the subscripts $\omega\theta$ can be dropped.

Equation (7) is just the normal-incidence radiation condition (absorbing boundary condition) modified by the addition of correction functions h_m which are obtained through solution of one-dimensional (1-D) wave equations along the boundary. In practice, just three terms ($M = 3$) in the series are sufficient to give reflection coefficients less than 0.01 for all real angles of incidence less than 89°, an enormous improvement over paraxial absorbing boundaries. In the immediate

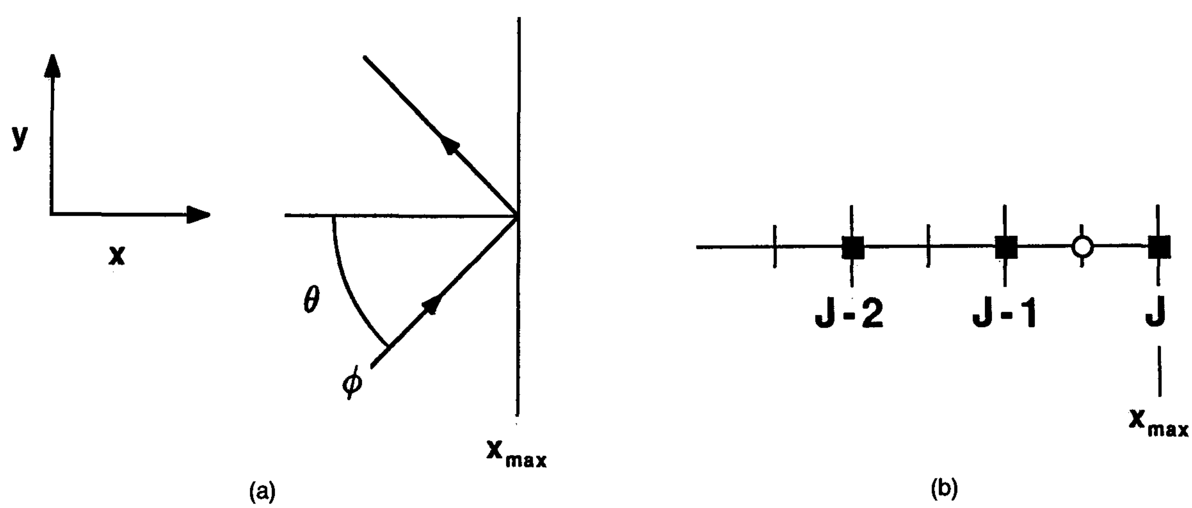

FIG. 1. (a) Wave incident on absorbing boundary $x = x_{\max}$ from the left. The angle of incidence is θ. (b) Finite-difference mesh nodes in the x direction. Displacements $\mathbf{u}$ are defined at $x_j = j\Delta x$ ($j = \cdots J-2,\ J-1,\ J$) (filled squares). Potentials ϕ, A_y, and A_z are defined at $x_{J\pm1/2}$ (open circles). A_x is defined at x_J.

neighborhood of $\theta = 90°$, the rational approximation must fail and the reflection coefficient becomes unity, as it also does for paraxial boundaries. Plots of the reflection coefficient are given by Lindman where the values for the coefficients α_m and β_m are also tabulated. These coefficients are obtained through minimization of a suitably weighted average of the reflection coefficient over the range of incidence angles up to $89°$.

In general, the sum over θ in equation (2) is not limited to real values. Complex values of θ corresponding to evanescent waves must often be accommodated. When the incident field contains significant energy in the evanescent portion of the spectrum, the boundary condition embodied in equations (7) and (8) is not sufficient for good absorption. In this case, a more general boundary condition (also given by Lindman) may be employed which absorbs both propagating and evanescent waves. To account for evanescent waves properly, the wave equations for the correction functions h_m must contain damping terms. The derivation is not reproduced here. The damping terms are proportional to $|k_y|$. No exact realization of this operator in the spatial (y) domain is available, so it is most convenient to apply this generalized boundary condition directly in the transform or wavenumber domain. The more general scalar absorbing boundary condition is

$$\frac{\partial\tilde{\phi}}{\partial t} + c\left(\sum_{m=1}^{m=M} \alpha_m\right)\frac{\partial\tilde{\phi}}{\partial x} = -\sum_{m=1}^{m=M} \tilde{h}_m, \tag{9}$$

where

$$\frac{\partial^2\tilde{h}_m}{\partial t^2} + c\gamma_m|k_y|\frac{\partial\tilde{h}_m}{\partial t} + c^2k_y^2\tilde{h}_m = \alpha_m\left[-c^2k_y^2 + (1-\gamma_m)c|k_y|\frac{\partial}{\partial t}\right]\left(c\frac{\partial\tilde{\phi}}{\partial x}\right), \tag{10}$$

and where a tilde denotes a transformed quantity. The coefficients α_m and γ_m are obtained by minimizing a weighted average of the reflection coefficients, which in this case contains contributions from both the propagating portion and the evanescent portion of the spectrum. This more general absorbing boundary condition gives excellent performance with six terms ($M = 6$) in the series. The reflection coefficient is again less than 0.01 throughout the range of real θ, except in the immediate neighborhood of $90°$. Moreover, the reflection coefficient is small for complex θ as well.

The transformations between the spatial and the wavenumber domains required for application of this boundary condition on a finite-difference mesh will, in most instances, be accomplished through the discrete Fourier transform. For problems in which the implied periodicity is inappropriate, an approximate representation of the operator $|k_y|$ as a filter operation in the spatial domain is given by Lindman.

AN ABSORBING BOUNDARY FOR ELASTIC WAVES

For the elastic wave equation, the field variable is the displacement vector $\mathbf{u} = (u_x, u_y, u_z)$. The relationship between a Fourier component of $\mathbf{u}$ and its normal gradient is given by a matrix rather than by a simple scalar. A scalar wave absorbing boundary condition cannot be applied directly to individual Cartesian components of displacement. Physically, the displacement components comprise both compressional and shear disturbances and, except at normal incidence, no unique propagation velocity can be ascribed to them. The matrix relating Fourier components of $\mathbf{u}$ and their gradients is quite complex; cf., equations (B-8) and (B-9). A straightforward application of the simple procedure outlined above is not a fruitful approach. Instead, it is much more efficient to employ intermediary quantities: the scalar and vector potentials.

Consider an isotropic, homogeneous, elastic medium with compressional and shear velocities c and s, respectively. The displacement field $\mathbf{u}$ satisfies the vector wave equation

$$\frac{\partial^2\mathbf{u}}{\partial t^2} = c^2\nabla(\nabla\cdot\mathbf{u}) - s^2\nabla\times\nabla\times\mathbf{u}, \tag{11}$$

and may be written as a sum of curl-free and divergence-free parts,

$$\mathbf{u} = \nabla\phi + \nabla\times\mathbf{A}, \tag{12}$$

where ϕ is the scalar (compressional) potential and $\mathbf{A}$ is the vector (shear) potential. The specification of the vector potential may be made unique through the auxilliary condition $\nabla\cdot\mathbf{A} = 0$. The potentials satisfy the wave equations

$$\frac{\partial^2\phi}{\partial t^2} = c^2\nabla^2\phi \tag{13}$$

and

$$\frac{\partial^2\mathbf{A}}{\partial t^2} = -s^2\nabla\times\nabla\times\mathbf{A} = s^2\nabla^2\mathbf{A}. \tag{14}$$

In view of equation (12), the right terms of equations (13) and (14) may be written in terms of the displacement vector $\mathbf{u}$:

$$\frac{\partial^2\phi}{\partial t^2} = c^2\nabla\cdot\mathbf{u} \tag{15}$$

and

$$\frac{\partial^2\mathbf{A}}{\partial t^2} = -s^2\nabla\times\mathbf{u}. \tag{16}$$

The compressional potential ϕ and the three Cartesian components of the shear potential $\mathbf{A}$ satisfy uncoupled scalar wave equations, and therefore may be individually treated by the scheme of Lindman. If $\mathbf{A}$ is initially divergence-free, it will remain so.

An absorbing boundary condition for finite-difference elastic wave calculations consisting of three steps may now be stated. First, equations (15) and (16) are used to express the vector displacement field in terms of four intermediary uncoupled scalar potentials, ϕ and the three Cartesian components of $\mathbf{A}$. Next, the scalar absorbing boundary condition embodied in equations (9) and (10) is applied to these potentials; and, finally, equation (12) is employed to recover the vector displacement field from the potentials. For all of the equations which define this three-step process, explicit, stable, second-order accurate finite-difference analogs have been obtained. In the development below, the finite-difference equations employed in the interior regions of the mesh are assumed to be the standard second-order accurate equations in the displacement formulation. For alternative differencing schemes such as the fourth-order accurate displacement formulation or a second-order or fourth-order accurate velocity-stress formulation (Virieux, 1986), similar results may be obtained.

The finite-difference equations of the interior mesh allow advancement of the displacement $\mathbf{u}$ from time $t^n = n\Delta t$ to $t^{n+1} = (n+1)\Delta t$ at all mesh points (x_j, y, z) for which $j < J$, where $x_j = j\Delta x$ and $x_J = x_{\max}$ is the boundary as shown in Figure 1b. The boundary condition must supply equations for $\mathbf{u}(x_J, y, z, t^{n+1}) = \mathbf{u}_J^{n+1}$. Potentials are defined as centered one-half cell inside the boundary (in the case of ϕ, A_y, and A_z) and on the boundary (in the case of A_x) and are advanced to time t^{n-1} through finite-difference analogs of equations (15) and (16).

$$\frac{\tilde{\phi}^{n+1}_{J-1/2} - 2\tilde{\phi}^n_{J-1/2} + \tilde{\phi}^{n-1}_{J-1/2}}{\Delta t^2} = c^2\left(\frac{\tilde{u}^n_{xJ} - \tilde{u}^n_{xJ-1}}{\Delta x} + ik_y\frac{\tilde{u}^n_{yJ} + \tilde{u}^n_{yJ-1}}{2} + ik_z\frac{\tilde{u}^n_{zJ} + \tilde{u}^n_{zJ-1}}{2}\right). \tag{17}$$

$$\frac{\tilde{A}^{n-1}_{xJ} - 2\tilde{A}^n_{xJ} + \tilde{A}^{n-1}_{xJ}}{\Delta t^2} = -s^2(ik_y\tilde{u}^n_{zJ} - ik_z\tilde{u}^n_{yJ}). \tag{18}$$

$$\frac{\tilde{A}^{n+1}_{yJ-1/2} - 2\tilde{A}^n_{yJ-1/2} + \tilde{A}^{n-1}_{yJ-1/2}}{\Delta t^2} = -s^2\left(ik_z\frac{\tilde{u}^n_{xJ} + \tilde{u}^n_{xJ-1}}{2} - \frac{\tilde{u}^n_{zJ} - \tilde{u}^n_{zJ-1}}{\Delta x}\right). \tag{19}$$

$$\frac{\tilde{A}^{n+1}_{zJ-1/2} - 2\tilde{A}^n_{zJ-1/2} + \tilde{A}^{n-1}_{zJ-1/2}}{\Delta t^2} = -s^2\left(\frac{\tilde{u}^n_{yJ} - \tilde{u}^n_{yJ-1}}{\Delta x} - ik_y\frac{\tilde{u}^n_{xJ} + \tilde{u}^n_{xJ-1}}{2}\right). \tag{20}$$

All quantities have been transformed into the wavenumber domain so that the scalar absorbing boundary conditions embodied in equations (9) and (10) can be applied to these scalar potentials. The simpler scheme using equations (7) and (8) is not sufficient because propagating shear waves incident on the boundary at angles greater than the critical angle, $\sin^{-1}(s/c)$, will excite evanescent compressional components along the boundary due to truncation error in the finite-difference scheme. If these evanescent components are not adequately handled by the boundary condition, numerical instability can develop.

In order to reconstruct time-advanced displacements on the boundary at x_J from equation (12), x derivatives of three of the potentials, $\tilde{\phi}$, $\tilde{A}_y$, and $\tilde{A}_z$ are required. These derivatives are approximated by finite differences across the boundary, i.e., differences between the values one-half cell inside the boundary $(x_{J-1/2})$ and those one-half cell outside the boundary $(x_{J+1/2})$. The former are obtained through equations (17)–(20); and the latter are obtained through application of the scalar absorbing boundary:

$$\frac{\tilde{\phi}^{n+1}_{J+1/2} - \tilde{\phi}^n_{J+1/2} + \tilde{\phi}^{n+1}_{J-1/2} - \tilde{\phi}^n_{J-1/2}}{2\Delta t} + cq\frac{\tilde{\phi}^{n+1}_{J+1/2} - \tilde{\phi}^{n+1}_{J-1/2} + \tilde{\phi}^n_{J+1/2} - \tilde{\phi}^n_{J-1/2}}{2\Delta x} = -\sum_{m=1}^{m=M}\tilde{h}^{n+1/2}_{\phi m}; \tag{21}$$

$$\frac{\tilde{A}^{n+1}_{yJ+1/2} - \tilde{A}^n_{yJ+1/2} + \tilde{A}^{n+1}_{yJ-1/2} - \tilde{A}^n_{yJ-1/2}}{2\Delta t} + sq\frac{\tilde{A}^{n+1}_{yJ+1/2} - \tilde{A}^{n+1}_{yJ-1/2} + \tilde{A}^n_{yJ+1/2} - \tilde{A}^n_{yJ-1/2}}{2\Delta x} = -\sum_{m=1}^{m=M}\tilde{h}^{n+1/2}_{ym}; \tag{22}$$

and

$$\frac{\tilde{A}^{n+1}_{zJ+1/2} - \tilde{A}^n_{zJ+1/2} + \tilde{A}^{n+1}_{zJ-1/2} - \tilde{A}^n_{zJ-1/2}}{2\Delta t} + sq\frac{\tilde{A}^{n+1}_{zJ+1/2} - \tilde{A}^{n+1}_{zJ-1/2} + \tilde{A}^n_{zJ+1/2} - \tilde{A}^n_{zJ-1/2}}{2\Delta x} = -\sum_{m=1}^{m=M}\tilde{h}^{n+1/2}_{zm}, \tag{23}$$

where $q = \sum \alpha_m$. Equations (21), (22), and (23) are solved for $\tilde{\phi}^{n+1}_{J+1/2}$, $\tilde{A}^{n+1}_{yJ+1/2}$, and $\tilde{A}^{n+1}_{zJ+1/2}$, respectively. In equation (21) the time-advanced correction function $\tilde{h}^{n+1/2}_{\phi m}$ is obtained through the finite-difference equation

$$\begin{aligned}&\frac{\tilde{h}^{n+1/2}_{\phi m} - 2\tilde{h}^{n-1/2}_{\phi m} + \tilde{h}^{n-3/2}_{\phi m}}{\Delta t^2} + c\gamma_m|k_\perp|\frac{\tilde{h}^{n+1/2}_{\phi m} - \tilde{h}^{n-3/2}_{\phi m}}{2\Delta t} + c^2k_\perp^2\tilde{h}^{n-1/2}_{\phi m}\\ &= -\alpha_m c^3k_\perp^2\frac{\tilde{\phi}^n_{J+1/2} - \tilde{\phi}^n_{J-1/2} + \tilde{\phi}^{n-1}_{J+1/2} - \tilde{\phi}^{n-1}_{J-1/2}}{2\Delta x}\\ &\quad + \alpha_m(1-\gamma_m)c^2|k_\perp|\frac{\tilde{\phi}^n_{J+1/2} - \tilde{\phi}^n_{J-1/2} - \tilde{\phi}^{n-1}_{J+1/2} + \tilde{\phi}^{n-1}_{J-1/2}}{\Delta t\Delta x},\end{aligned} \tag{24}$$

where $k_\perp^2 = (k_y^2 + k_z^2)$. Analogous equations apply for $\tilde{h}^{n+1/2}_{ym}$ and $\tilde{h}^{n+1/2}_{zm}$ (with s replacing c and $\tilde{A}_y$ or $\tilde{A}_z$ replacing $\tilde{\phi}$.)

Finally, time-advanced displacements $\tilde{\mathbf{u}}_J^{n+1}$, on the boundary $x_J = J\Delta x$, are obtained through a finite-difference analog of equation (12) applied to the time-advanced potentials:

$$\tilde{u}^{n+1}_{xJ} = \frac{\tilde{\phi}^{n+1}_{J+1/2} - \tilde{\phi}^{n+1}_{J-1/2}}{\Delta x} + ik_y\frac{\tilde{A}^{n+1}_{zJ+1/2} + \tilde{A}^{n+1}_{zJ-1/2}}{2} - ik_z\frac{\tilde{A}^{n+1}_{yJ+1/2} + \tilde{A}^{n+1}_{yJ-1/2}}{2}; \tag{25}$$

$$\tilde{u}^{n+1}_{yJ} = ik_y\frac{\tilde{\phi}^{n+1}_{J+1/2} + \tilde{\phi}^{n+1}_{J-1/2}}{2} - \frac{\tilde{A}^{n+1}_{zJ+1/2} - \tilde{A}^{n+1}_{zJ-1/2}}{\Delta x} + ik_z\tilde{A}^{n+1}_{xJ}; \tag{26}$$

and

$$\tilde{u}^{n+1}_{zJ} = ik_z\frac{\tilde{\phi}^{n+1}_{J+1/2} + \tilde{\phi}^{n+1}_{J-1/2}}{2} + \frac{\tilde{A}^{n+1}_{yJ+1/2} - \tilde{A}^{n+1}_{yJ-1/2}}{\Delta x} - ik_y\tilde{A}^{n+1}_{xJ}. \tag{27}$$

Solution of these equations, followed by an inverse Fourier transform to the spatial domain, completes the absorbing boundary condition.

A comment on the definition of the transverse wavenumbers k_y and k_z employed in the foregoing equations is appropriate. In the boundary condition, the transverse derivatives $\partial/\partial y$ and

$\partial/\partial z$ have been replaced with the exact Fourier-domain representations ik_y and ik_z. This causes a slight "mismatch" between the boundary condition and interior mesh finite-difference equations in which derivatives are only approximate because of truncation error. This mismatch, which results in a small degradation of the performance of the boundary condition, can be reduced by appropriate definitions of k_y and k_z. For example, if the interior finite-difference equations are accurate to second order, then replacing k_y by $k'_y = k_y \operatorname{sinc}(k_y \Delta y/2)$ and k_z by $k'_z = k_z \operatorname{sinc}(k_z \Delta z/2)$ in equations (17)–(27) eliminates this mismatch to a great extent. With reasonable finite-difference grid resolution ($k_y \Delta y$, $k_z \Delta z \le 0.5$), the primed and unprimed wavenumbers will differ by 1 percent or less. For simplicity of notation, a complex Fourier transform has been employed for the transformation between the spatial and wavenumber domains. In most applications, of course, a real Fourier transform would be sufficient, resulting in very simple modifications to the foregoing equations.

Finally, it is well known that solutions of the vector wave equation in three dimensions may be expressed in terms of three potentials ϕ, ψ, and χ, each satisfying a scalar wave equation (Morse and Feshbach, 1953), whereas the new boundary condition uses four scalar potentials. Only three of these potentials, ϕ, A_y, and A_z, however, require the scalar absorbing boundary condition to be invoked. The fourth, A_x, may be considered an auxiliary function in that a boundary condition is not required for it. However, the extra degree of freedom conferred by this additional potential allows an extremely simple transformation between the potentials and the Cartesian components of the displacement, namely equations (12), (15), and (16). In a three-potential scheme, the transformation between potentials and displacements is far more complex, cf., equations (B-5)–(B-7); and the existence of finite-difference equations analogous to equations (17)–(20) and (25)–(27) is not apparent.

THE REFLECTIVITY MATRIX $\underset{\sim}{R}$

A quantitative measure of the performance of absorbing boundary conditions is provided by the reflectivity matrix $\underset{\sim}{R}$ which relates the reflected vector displacement field to the incident field. In this section, the reflectivity matrix of the new absorbing boundary condition, including finite-difference truncation error, is determined.

The absorbing boundary condition has been formulated in the time-wavenumber domain. Reflectivities are properly considered in the frequency-wavenumber domain. Transforming to the frequency domain and defining the four component vectors

$$\tilde{\mathbf{\Phi}}_{J-1/2} = (\tilde{\phi}_{J-1/2}, \tilde{A}_{xJ}, \tilde{A}_{yJ-1/2}, \tilde{A}_{zJ-1/2}) \tag{28}$$

$$\tilde{\mathbf{\Phi}}_{J+1/2} = (\tilde{\phi}_{J+1/2}, \tilde{A}_{xJ}, \tilde{A}_{yJ+1/2}, \tilde{A}_{zJ+1/2}), \tag{29}$$

equations (17)–(20), (21)–(23), and (25)–(27) may be written in vector form:

$$\tilde{\mathbf{\Phi}}_{J-1/2} = \underset{\sim}{A}\tilde{\mathbf{u}}_{J-1} + \underset{\sim}{B}\tilde{\mathbf{u}}_J; \tag{30}$$

$$\tilde{\mathbf{\Phi}}_{J+1/2} = (\underset{\sim}{I} + \underset{\sim}{C})\tilde{\mathbf{\Phi}}_{J-1/2}; \tag{31}$$

and

$$\tilde{\mathbf{u}}_J = \underset{\sim}{D}\tilde{\mathbf{\Phi}}_{J-1/2} + \underset{\sim}{E}\tilde{\mathbf{\Phi}}_{J+1/2}, \tag{32}$$

where $\underset{\sim}{I}$ is the identity matrix and $\underset{\sim}{A}$, $\underset{\sim}{B}$, $\underset{\sim}{C}$, $\underset{\sim}{D}$, and $\underset{\sim}{E}$ are matrices given in Appendix A. Solving for $\tilde{\mathbf{u}}_J$,

$$\tilde{\mathbf{u}}_J = (\underset{\sim}{I} + \underset{\sim}{G})\tilde{\mathbf{u}}_{J-1}, \tag{33}$$

where

$$\underset{\sim}{G} = \left\{\underset{\sim}{I} - \left[\underset{\sim}{D} + \underset{\sim}{E}(\underset{\sim}{I} + \underset{\sim}{C})\right]\underset{\sim}{B}\right\}^{-1} \times \left\{\left[\underset{\sim}{D} + \underset{\sim}{E}(\underset{\sim}{I} + \underset{\sim}{C})\right](\underset{\sim}{A} + \underset{\sim}{B}) - \underset{\sim}{I}\right\}. \tag{34}$$

In order to define reflectivity, the displacement field $\tilde{\mathbf{u}}$ is decomposed into incident (+) and reflected (−) parts:

$$\tilde{\mathbf{u}} = \tilde{\mathbf{u}}_+ + \tilde{\mathbf{u}}_-. \tag{35}$$

The reflectivity matrix $\underset{\sim}{R}$ assumes its simplest and most readily interpreted form when the incident and reflected fields are expressed in terms of the normal modes of the system, which in this case are the normal modes of the finite-difference equations employed in the interior region of the mesh. Let the transformations be denoted by

$$\tilde{\mathbf{U}}_\pm = \underset{\sim}{M}_\pm \tilde{\mathbf{u}}_\pm, \tag{36}$$

where $\tilde{\mathbf{u}}_\pm$ and $\tilde{\mathbf{U}}_\pm$ are the coordinates of displacement with respect to the Cartesian unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, and with respect to the vectors corresponding to normal modes, respectively. The $\underset{\sim}{M}_\pm$ are transformation matrices which describe the change of basis. The normal modes, by definition, propagate in the x direction as simple exponentials so that

$$\tilde{\mathbf{U}}_\pm(x) = \tilde{\mathbf{U}}_\pm(0)\exp(\pm i\underset{\sim}{K}x), \tag{37}$$

where $\underset{\sim}{K}$ is a diagonal matrix whose nonzero elements K_{ii} are the roots of the finite-difference dispersion relation, equation (A-12). Then equation (33) may be written

$$\underset{\sim}{M}_+^{-1}\tilde{\mathbf{U}}_+ + \underset{\sim}{M}_-^{-1}\tilde{\mathbf{U}}_- = (\underset{\sim}{I} + \underset{\sim}{G})\left[\underset{\sim}{M}_+^{-1}\exp(-i\underset{\sim}{K}\Delta x)\tilde{\mathbf{U}}_+ + \underset{\sim}{M}_-^{-1}\exp(i\underset{\sim}{K}\Delta x)\tilde{\mathbf{U}}_-\right]. \tag{38}$$

Solving for $\tilde{\mathbf{U}}_-$ and defining the reflectivity matrix through

$$\tilde{\mathbf{U}}_- = \underset{\sim}{R}\tilde{\mathbf{U}}_+, \tag{39}$$

an expression for $\underset{\sim}{R}$ is obtained:

$$\underset{\sim}{R} = \left[(\underset{\sim}{I} + \underset{\sim}{M}_-\underset{\sim}{G}\underset{\sim}{M}_-^{-1})\exp(i\underset{\sim}{K}\Delta x) - \underset{\sim}{I}\right]^{-1}\left[\underset{\sim}{M}_-\underset{\sim}{M}_+^{-1}\right] \times \left[\underset{\sim}{I} - (\underset{\sim}{I} + \underset{\sim}{M}_+\underset{\sim}{G}\underset{\sim}{M}_+^{-1})\exp(-i\underset{\sim}{K}\Delta x)\right]. \tag{40}$$

Before describing the numerical results of the reflectivity calculations, it is instructive to consider briefly the three normal modes of the isotropic elastic wave equation in three dimensions. The first mode is a compressional mode with displacement parallel to the propagation vector $\mathbf{k}$:

$$\tilde{\mathbf{u}}_{c\pm} = \pm k_{xc}\hat{\mathbf{x}} + k_y\hat{\mathbf{y}} + k_z\hat{\mathbf{z}}, \tag{41}$$

where $k_{xc}^2 = (k_c^2 - k_\perp^2)$, $k_\perp^2 = (k_y^2 + k_z^2)$, and $k_c = \omega/c$. The remaining two modes are the two orthogonal polarizations of shear modes, which have displacement vectors perpendicular to $\mathbf{k}$:

$$\tilde{\mathbf{u}}_{1\pm} = \mp k_\perp \hat{\mathbf{x}} + \frac{k_y k_{xs}}{k_\perp} \hat{\mathbf{y}} + \frac{k_z k_{xs}}{k_\perp} \hat{\mathbf{z}} \tag{42}$$

and

$$\tilde{\mathbf{u}}_{2\pm} = \frac{k_s k_z}{k_\perp} \hat{\mathbf{y}} - \frac{k_s k_y}{k_\perp} \hat{\mathbf{z}}, \tag{43}$$

where $k_{xs}^2 = (k_s^2 - k_\perp^2)$ and $k_s = \omega/s$. The dispersion equation has single roots at $k_x = \pm k_{xc}$ and double roots at $k_x = \pm k_{xs}$, so that $\underset{\sim}{\mathbf{K}}$ has the diagonal elements $K_{ii} = (k_{xc}, k_{xs}, k_{xs})$, where the positive roots are chosen. The modes have been normalized to the magnitude of the compressional or shear wavenumbers k_c or k_s. The displacement vectors $\tilde{\mathbf{u}}_\pm$ may be expanded in terms of these normal modes

$$\tilde{\mathbf{u}}_\pm = U_{c\pm} \tilde{\mathbf{u}}_{c\pm} + U_{1\pm} \tilde{\mathbf{u}}_{1\pm} + U_{2\pm} \tilde{\mathbf{u}}_{2\pm}. \tag{44}$$

The expansion coefficients $U_{c\pm}$, $U_{1\pm}$, and $U_{2\pm}$ are in fact the three components of $\mathbf{U}_\pm$. From equation (44) it follows that the transformation matrices $\underset{\sim}{\mathbf{M}}_\pm$ may be formed from the column vectors representing the components of the three normal modes:

$$\underset{\sim}{\mathbf{M}}_\pm = \{\tilde{\mathbf{u}}_{c\pm}, \tilde{\mathbf{u}}_{1\pm}, \tilde{\mathbf{u}}_{2\pm}\}^{-1}. \tag{45}$$

For the finite-difference analog of the elastic wave equation, the roots of the dispersion relation and the corresponding normal modes must be found numerically. They differ from the results given above because of the influence of truncation error. For example, the finite-difference mode corresponding to $\mathbf{u}_{2\pm}$ has a small but nonzero x component, and the two shear modes have slightly different wavenumbers (propagation speeds.) The matrices $\underset{\sim}{\mathbf{K}}$, $\underset{\sim}{\mathbf{M}}_\pm$, and $\underset{\sim}{\mathbf{G}}$ required for the calculation of the reflectivity matrix $\underset{\sim}{\mathbf{R}}$ are given in Appendix A.

NUMERICAL RESULTS

In Figure 2 the reflectivity matrix of the new boundary condition is compared with that of the second-order paraxial boundary condition. For simplicity, finite-difference truncation error has been made negligible and the propagation is two-dimensional; $k_z = u_z = 0$. A shear to compressional velocity ratio, $s/c = 1/2$, is employed (Poisson's ratio of 1/3). The reflectivity matrix has four entries: R_{cc} and R_{cs}, the reflectivities for an incident compressional wave reflected as compressional and shear, respectively; and R_{sc} and R_{ss}, the reflectivities for an incident shear wave reflected as compressional and shear, respectively. The absolute values of R_{cc} and R_{cs} are plotted in Figure 2a versus the compressional wave incident angle $\theta_c = \sin^{-1}(ck_y/\omega)$, measured in degrees. (All numerical results are displayed in terms of absolute values of reflection coefficients.) The new boundary condition isolates the compressional and shear waves exactly via calculation of the potential. Therefore, the reflectivity matrix $\underset{\sim}{\mathbf{R}}$ is diagonal and the diagonal elements are simply the reflection coefficients of Lindman's scalar absorbing boundary condition, equations (9) and (10). R_{cc} for this new boundary condition (labeled L in the figure) is extremely small for all incident angles less than about 89°. In contrast, the compressional wave reflectivities (particularly R_{cc}) of the paraxial boundary condition, labeled CC and CS in Figure 2, become quite large beyond about 60°. Quantitative comparison of the two absorbing boundaries is facilitated by plotting the common logarithms of the reflectivities in Figure 2b. In Figures 2c and 2d the shear wave reflectivities of both the new boundary condition and the paraxial absorbing boundary condition are displayed. The new boundary condition again displays no cross reflection (R_{sc} = 0), because of the complete separation of shear and compressional waves, and the shear-shear reflectivity R_{ss} (labeled L in the figure) is again simply the very low reflectivity of the scalar absorbing boundary. For incident shear waves the paraxial approximation begins to fail for angles $\theta_s = \sin^{-1}(sk_y/\omega)$ of less than 30°. The discontinuity in slope of the paraxial coefficients at exactly 30° is due to the formation of a "pseudo-head wave" (evanescent compressional component) along the boundary. Clearly, for negligible finite-difference truncation error, the new boundary condition is one to two orders of magnitude less reflective than the second-order paraxial scheme for both incident compressional waves and shear waves throughout the range of large incidence angles (except for the immediate neighborhood of 90°, where both schemes fail).

The reflectivity of the new boundary condition, again in two dimensions, is calculated and compared in Figure 3 for two typical levels of truncation error. The compressional and shear speeds are $c = 1$ and $s = 0.5$, and the finite-difference grid parameters are $\Delta x = \Delta y = 0.15$ and $\Delta t = 0.075$. Plots of the four reflection coefficients of the new absorbing boundary are displayed for three frequencies, $\omega \approx 0$, $\omega = 1$, and $\omega = 2$. For $\omega \approx 0$, truncation error is negligible and the curves are reproduced from Figures 2b and 2d. For $\omega = 1$ (about 40 and 20 grid points per compressional and shear wavelength, respectively), truncation error is moderate; while for $\omega = 2$ (about 20 and 10 grid points per wavelength), truncation error is quite large. The reflectivity curves R_{cc} and R_{ss} are qualitatively similar for the three levels of grid resolution. As truncation error is increased, the reflectivities at small and intermediate angles of incidence are somewhat increased. The width of the poor performance region around 90°, where the reflectivity is dominated by the errors in the rational expansion employed in the scalar absorbing boundary itself, is little affected. The shear-shear reflectivity R_{ss} is the more sensitive to truncation error, since shear waves are relatively less well resolved than compressional waves at a given frequency.

The cross reflectivities for the new boundary condition R_{cs} and R_{sc} take on nonzero values when finite-difference truncation error is introduced. Compressional and shear disturbances are no longer exactly separated by the finite-difference potentials. R_{cs} remains quite small even at low grid resolution, while R_{sc} is relatively larger, attaining values through the intermediate incidence angle region of about 0.03 and 0.12 for cases of moderate and large truncation error, respectively. In an extremely small neighborhood around the critical angle $\sin^{-1}(s/c)$, R_{sc} rises to a value of 1. However, the width of the significant reflectivity peak is so small (a fraction of a degree for the moderate truncation error case) that the integrated reflectivity under the peak remains negligible.

In Figures 4a–4c, reflectivities for full 3-D propagation are displayed. The elastic parameters chosen are $c = 1$ and $s = 0.577$ (Poisson's ratio of 1/4). The time and spatial dis-

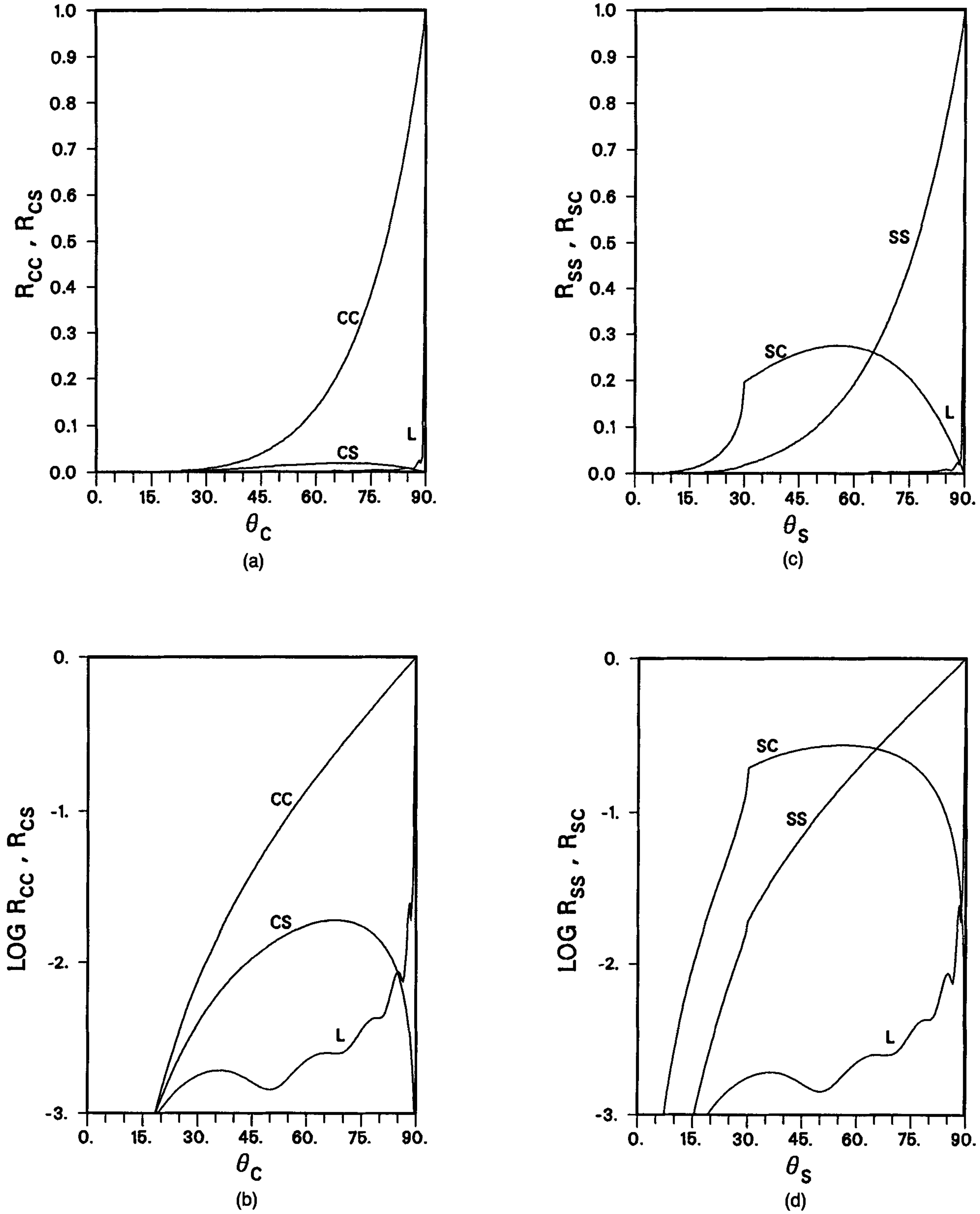

FIG. 2. Reflectivities for absorbing boundaries in two dimensions neglecting finite-difference truncation error. The shear-to-compressional speed ratio is 1/2 (Poisson's ratio of 1/3). (a) Curves *CC* and *CS* are paraxial approximations R_{cc} and R_{cs}, respectively, versus compressional wave incidence angle θ_c. Curve *L* is R_{cc} for the new boundary condition which uses Lindman's scheme. $R_{cs} = 0$ for the new boundary condition without truncation error. (b) Common logarithm of reflectivities in (a). (c) Curves *SS* and *SC* are paraxial approximations R_{ss} and R_{sc}, respectively, versus shear-wave incidence angle θ_s. Curve *L* is R_{ss} for the new boundary condition. $R_{sc} = 0$ for the new boundary condition without truncation error. (d) Common logarithm of reflectivities in (c).

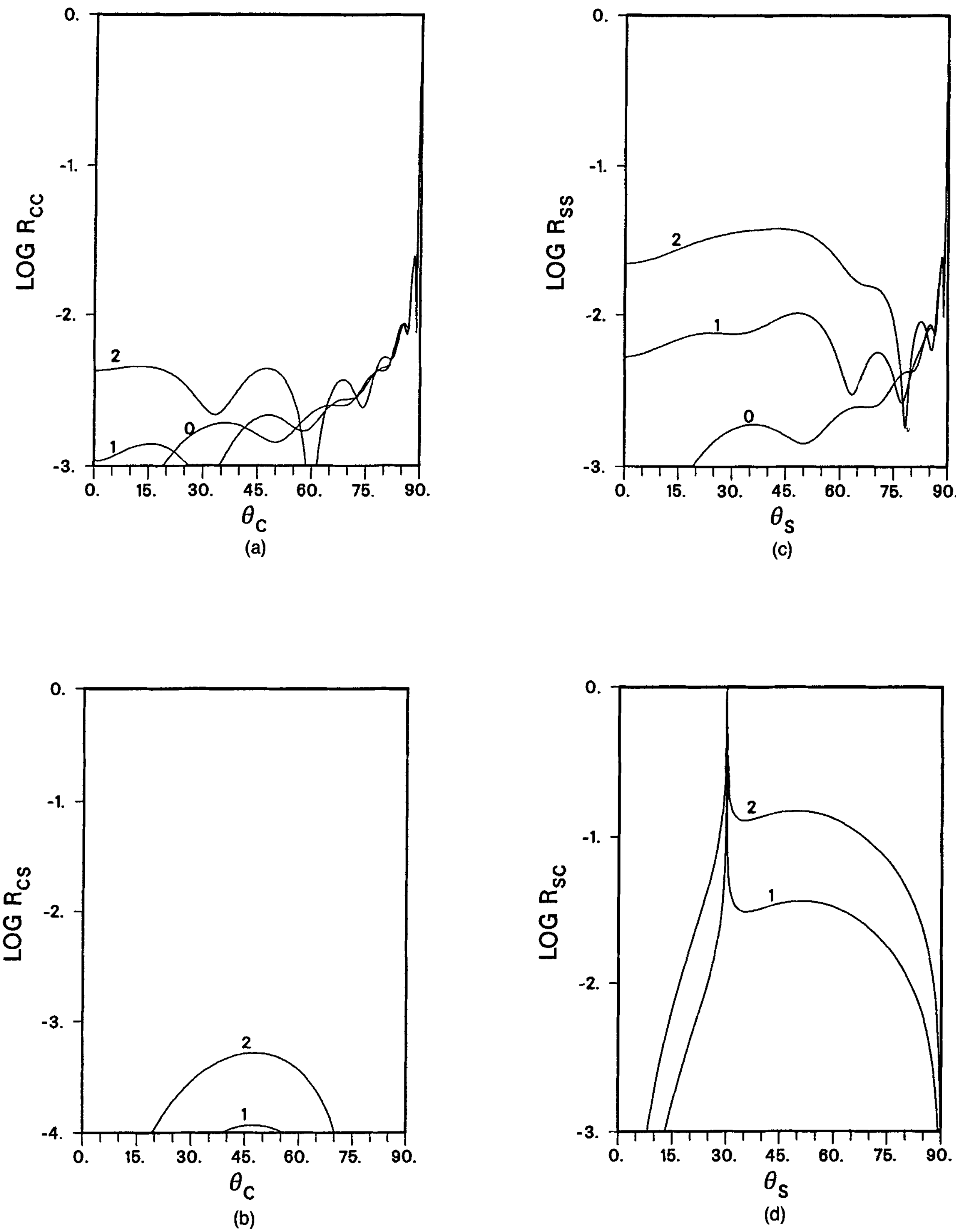

FIG. 3. Reflectivities for the new absorbing boundary condition in two dimensions including the effects of finite-difference truncation error. Parameters are $c = 1.$, $s = 0.5$, $\Delta x = \Delta y = 0.15$, and $\Delta t = 0.075$. Curves 0, 1, and 2 are for $\omega = 0$ (negligible truncation error), $\omega = 1$ (modest truncation error), and $\omega = 2$ (large truncation error). (a) Common logarithm of R_{cc} versus compressional incidence angle. (b) Common logarithm of R_{cs} versus compressional incidence angle. (c) Common logarithm of R_{ss} versus shear incidence angle. (d) Common logarithm of R_{sc} versus shear incidence angle.

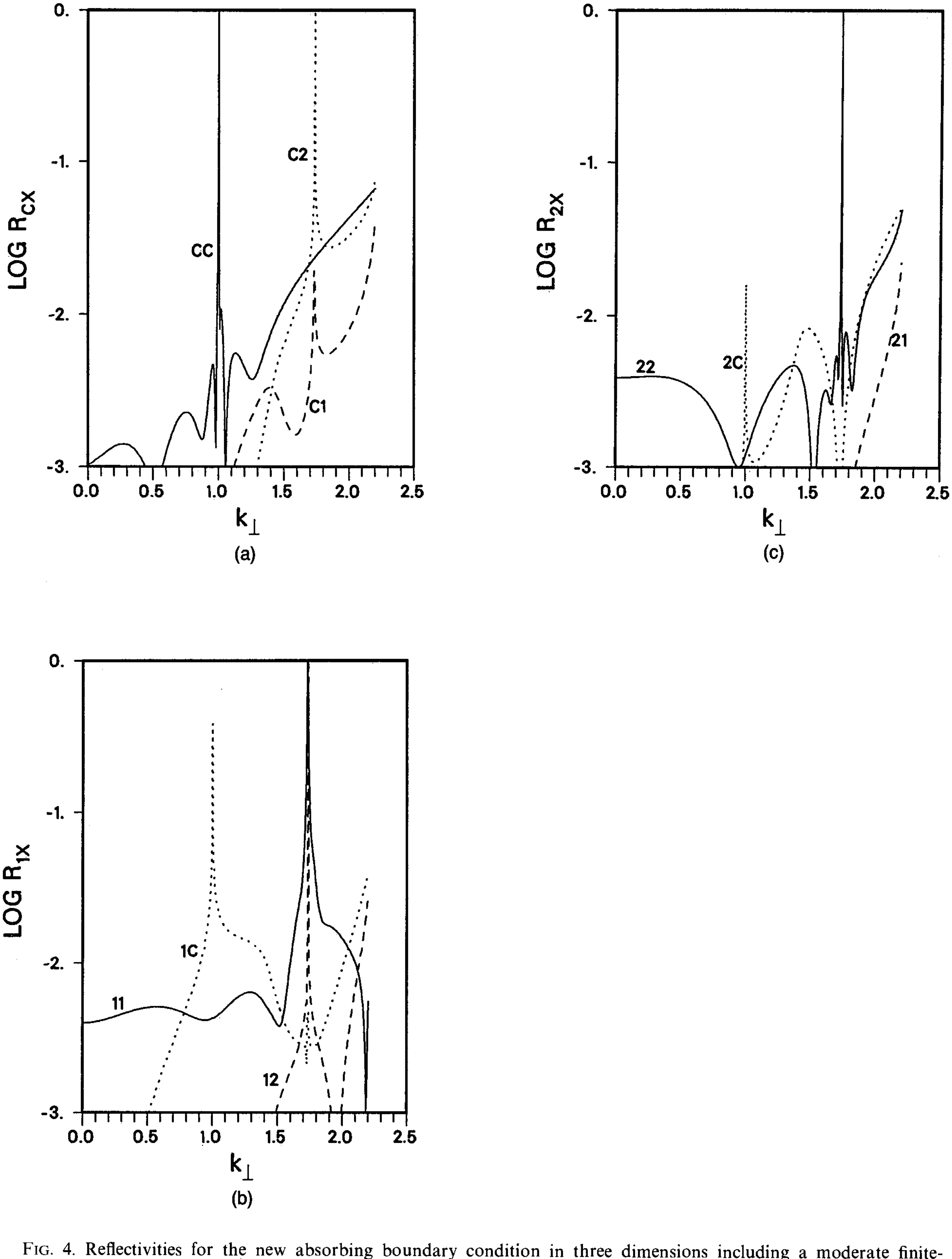

FIG. 4. Reflectivities for the new absorbing boundary condition in three dimensions including a moderate finite-difference truncation error. Parameters are $c = 1.$, $s = 0.577$, $\Delta x = \Delta y = \Delta z = 0.15$, $\Delta t = 0.075$, and $\omega = 1$. (a) Reflectivities for an incident compressional wave. Curves CC, $C1$, and $C2$ are reflectivities into compressional (R_{cc}), into shear wave of polarization 1 (R_{c1}), and into shear wave of polarization 2 (R_{c2}), respectively. (b) Reflectivities for an incident shear wave of polarization 1. Curves $1C$, 11, and 12 are reflectivities into compressional (R_{1c}), into shear wave of polarization 1 (R_{11}), and into shear wave of polarization 2 (R_{12}), respectively. (c) Reflectivities for an incident shear wave of polarization 2. Curves $2C$, 21, and 22 are reflectivities into compressional (R_{2c}), into shear wave of polarization 1 (R_{21}), and into shear wave of polarization 2 (R_{22}), respectively.

cretization remain unchanged and the intermediate grid resolution $\omega = 1$ is chosen. In Figures 4 the common logarithms of the reflectivities are plotted versus the transverse wavenumber $k_\perp = (k_y^2 + k_z^2)^{1/2}$. A constant ratio of $k_y^2/k_z^2 = 1.5$ is maintained as $k_\perp$ is varied. The label C stands for the compressional normal mode, and the labels 1 and 2, for the two polarizations of shear modes, as defined in the previous section. Plots of the three reflectivities R_{cc}, R_{c1}, and R_{c2}, corresponding to reflection of the incident compressional mode into compressional, shear 1, and shear 2 modes, are displayed in Figure 4a. The behavior of R_{cc} is similar to that in Figure 3a for the 2-D case in the region corresponding to propagating compressional waves, $k_\perp \leq k_c = 1$. In Figure 4a, however, the excellent absorption of the new boundary condition for evanescent waves, $k_\perp > k_c = 1$, is also evident. The high-reflectivity region is restricted to an extremely narrow spike at grazing incidence. The cross-reflection coefficients R_{c1} and R_{c2} become visible on the plot as $k_\perp$ increases beyond k_c, and display very narrow spikes near the grazing incidence point for shear waves, $k_\perp = k_s \approx 1.73$. For even larger values of $k_\perp$ beyond the grazing incidence angle for shear, the reflectivity increases due to increasing finite-difference truncation error. However, the reflectivity at very large $k_\perp$ is of little practical consequence, since both compressional and shear waves are highly evanescent in this region. For example, at $k_\perp = 2.2$ and for the parameters of Figure 4, compressional and shear wave amplitudes drop by about an order of magnitude in just 8 and 11 grid points, respectively, in the x direction.

The remaining six reflection coefficients (for either of the two shear modes 1 or 2 incident on the absorbing boundary) are displayed in Figures 4b and 4c. In these two cases, complete reflection of the incident normal mode into itself takes place within an extremely narrow region near $k_\perp \approx k_s$, while a narrow spike in the cross reflectivity into the compressional mode occurs at $k_\perp \approx k_c$. For all other perpendicular wavenumbers, the reflectivities are a few percent or less.

Up to this point, the performance of the new boundary condition and the comparison to the second-order paraxial boundary condition have been observed within the context of reflectivity matrices. In Figure 5, "snapshots" from 2-D elastic finite-difference calculations are displayed, demonstrating the very low reflectivity of the new boundary condition. A point force in the x direction is applied at the origin with a time dependence in the form of a wide-band zero-mean pulse. The center frequency of the time pulse is $\omega = 1$. The parameters $c = 1$, $s = 0.5$, $\Delta x = \Delta y = 0.15$, and $\Delta t = 0.075$ from Figure 3 are retained. Figure 5a shows the results when the new absorbing boundary conditions are employed at the upper and

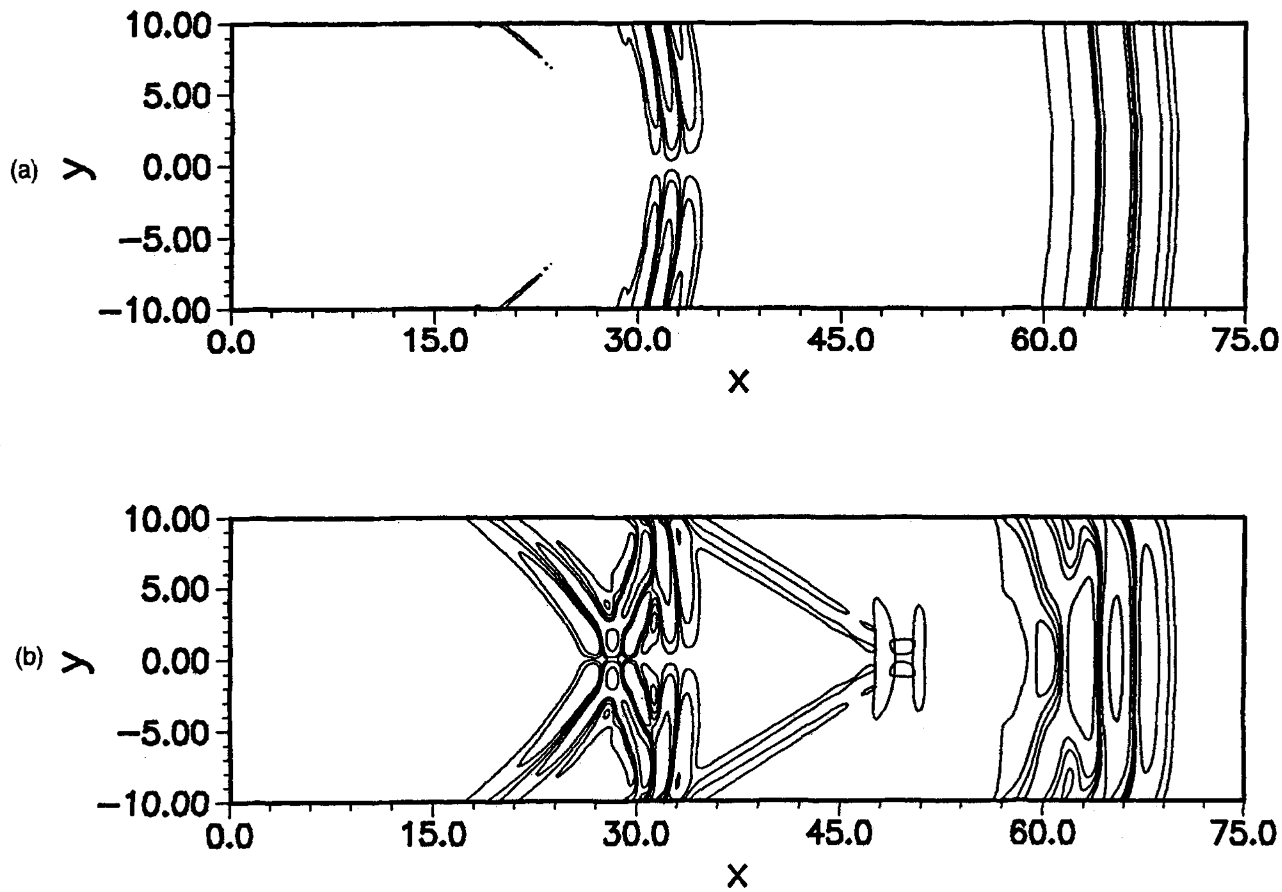

FIG. 5. Contours of $|\nabla \cdot \mathbf{u}|$ and $|\nabla \times \mathbf{u}|$ for 2-D finite-difference calculations with a horizontal point force at the origin. The lowest contour level is 2 percent of the maximum amplitude. (a) New absorbing boundary condition along the upper and lower boundaries. (b) Second-order paraxial absorbing boundary condition along the upper and lower boundaries.

lower boundaries $y = \pm 10$. The clusters of lines centered near $x = 64$ are contours of $|\nabla \cdot \mathbf{u}|$ (compressional wavelet) and those near $x = 32$ are contours of $|\nabla \times \mathbf{u}|$ (shear wavelet) at time $t = 72$. The behavior shown in Figure 5 for a compressional wavelet expanding cylindrically with nulls along the line $x = 0$ and, following behind, a cylindrically expanding shear wavelet with nulls along the line $y = 0$ agrees with propagation theory. The effective angles of incidence of the compressional and shear wavefronts with respect to the absorbing boundaries at the time of this plot are about 82° and 73°, respectively. The lowest contour level displayed is 2 percent of the maximum. No reflection or mode conversion is apparent at this contour level for the compressional component, consistent with reflection coefficient calculations in Figures 3a and 3b. For the shear component, small reflected wavelets (slightly less than 3 percent of the maximum) appear on the upper and lower boundaries near $x = 20$. Ray arguments show that these reflected shear wavelets were produced along the boundaries in the region near $x = 7$ to 10, where the local angle of incidence is about 30° to 45°, and that they then propagated across the mesh to their present positions. The shear-shear reflection coefficient plotted on Figure 3c shows a broad maximum in this range of angles. As expected from Figure 3c, the frequency of the reflected wavelets is much higher than that of the incident wavelet.

In contrast to the extremely small reflections displayed by the new boundary condition, very strong reflections are apparent in Figure 5b, the corresponding plot when second-order paraxial absorbing boundary conditions are employed. The compressional wavelet is highly distorted by multiple reflections, and a pseudo-head wave is generated. Large reflections are also apparent for the shear wavelet.

Waveforms from these two finite-difference calculations are displayed in Figures 6a and 6b. The x displacement at $(x, y) = (20, 0)$ is shown in Figure 6a as a function of time for the new absorbing boundary (solid line) and for the paraxial boundary (dotted line.) Also plotted as a solid line is the analytical result (Graff, 1975). The first signal to arrive at $t = 20$ is the compressional wave, followed at $t = 40$ by the shear contribution with a lower amplitude. The finite-difference results with the new boundary and the analytical results are essentially identical except for the very low-amplitude, high-frequency shear-shear reflection near $t = 55$. The paraxial boundary result displays several large-amplitude spurious signals due to boundary reflections. In Figure 6b the corresponding waveforms at $(x, y) = (50, 0)$ are shown. In this case only the compressional component arrives within the plotted time interval. Again the finite-difference result employing the new boundary condition and the analytical result are nearly identical. Because of the very large effective angle of incidence, the paraxial results are grossly in error due to spurious boundary reflectivity.

In the finite-difference example above, the Poisson's ratio is a nominal 1/3 ($s/c = 1/2$). However, numerical experiments have shown the new algorithm to be stable for any physical Poisson's ratio $0 \leq \nu \leq 1/2$. So long as the grid resolves the shear waves, the results remain accurate as s/c becomes small.

The cost of the new boundary condition in terms of compu-

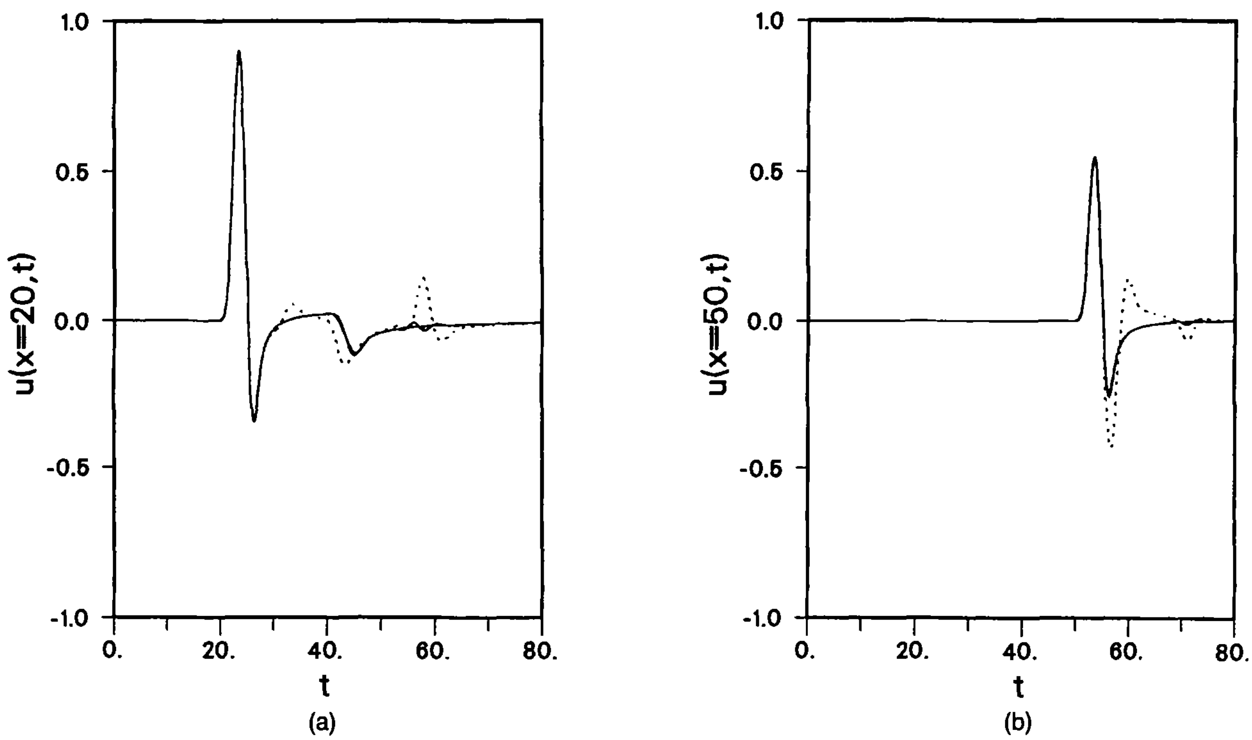

FIG. 6. Horizontal displacement versus time for the finite-difference calculations of Figures 5a and 5b at two positions along the x-axis. The solid curves are for calculations employing the new absorbing boundary conditions. The dotted curves are for second-order paraxial boundaries. Also plotted as solid curves are analytical results. (a) Horizontal displacement versus time at $x = 20$. (b) Horizontal displacement versus time at $x = 50$.

tation time will, of course, vary with the particular realization of the algorithm and the architecture of the computer employed. However, for 2-D, second-order accurate calculations the operation count is roughly equivalent to about 15 additional grid rows in the transverse (y) direction.

CONCLUSION

A new absorbing boundary condition has been given for finite-difference solutions to the elastic wave equation. The boundary condition uses the notion of scalar potentials to separate compressional and shear components of the incident vector displacement field. The elegant, efficient scheme of Lindman for scalar waves may then be applied to the scalar components. The reflectivity matrix of the new boundary condition has been calculated as a function of elastic medium parameters and finite-difference discretization. The new boundary has been found to be much more absorptive than the second-order paraxial absorbing boundary in general use and is numerically stable for any physical Poisson's ratio.

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APPENDIX A

MATRICES USED IN REFLECTIVITY CALCULATION

Transformed to the frequency domain and with the notation defined in equations (28)–(29), equations (17)–(20) become

$$\tilde{\mathbf{\Phi}}_{J-1/2} = \underset{\sim}{\mathbf{A}}\tilde{\mathbf{u}}_{J-1} + \underset{\sim}{\mathbf{B}}\tilde{\mathbf{u}}_J, \tag{A-1}$$

where $\underset{\sim}{\mathbf{A}}$ and $\underset{\sim}{\mathbf{B}}$ are the 4×3 matrices

$$\begin{Bmatrix} \underset{\sim}{\mathbf{A}} \\ \underset{\sim}{\mathbf{B}} \end{Bmatrix} = \frac{-\Delta x}{8\sin^2\left(\dfrac{\omega\Delta t}{2}\right)} \times \begin{bmatrix} \mp 2\beta_c^2 & i\beta_c^2 k_y \Delta x & i\beta_c^2 k_z \Delta x \\ \begin{Bmatrix} 0 \\ 0 \end{Bmatrix} & \begin{Bmatrix} 0 \\ 2i\beta_s^2 k_z \Delta x \end{Bmatrix} & \begin{Bmatrix} 0 \\ -2i\beta_s^2 k_y \Delta x \end{Bmatrix} \\ -i\beta_s^2 k_z \Delta x & 0 & \mp 2\beta_s^2 \\ i\beta_s^2 k_y \Delta x & \pm 2\beta_s^2 & 0 \end{bmatrix}. \tag{A-2}$$

The upper and lower signs apply for $\underset{\sim}{\mathbf{A}}$ and $\underset{\sim}{\mathbf{B}}$, respectively; and $\beta_c = c\Delta t/\Delta x$ and $\beta_s = s\Delta t/\Delta x$. Similarly, equations (25)–(27) become

$$\tilde{\mathbf{u}}_J = \underset{\sim}{\mathbf{D}}\tilde{\mathbf{\Phi}}_{J-1/2} + \underset{\sim}{\mathbf{E}}\tilde{\mathbf{\Phi}}_{J+1/2}, \tag{A-3}$$

where $\underset{\sim}{\mathbf{D}}$ and $\underset{\sim}{\mathbf{E}}$ are the 3×4 matrices

$$\begin{Bmatrix} \underset{\sim}{\mathbf{D}} \\ \underset{\sim}{\mathbf{E}} \end{Bmatrix} = \frac{1}{2\Delta x} \begin{bmatrix} \mp 2 & 0 & -ik_z \Delta x & ik_y \Delta x \\ ik_y \Delta x & ik_z \Delta x & 0 & \pm 2 \\ ik_z \Delta x & -ik_y \Delta x & \mp 2 & 0 \end{bmatrix}. \tag{A-4}$$

The finite-difference analogs of the scalar Lindman absorbing boundary condition, equations (21)–(23), become

$$\tilde{\mathbf{\Phi}}_{J+1/2} = (\underset{\sim}{\mathbf{I}} + \underset{\sim}{\mathbf{C}})\tilde{\mathbf{\Phi}}_{J-1/2}, \tag{A-5}$$

where $\underset{\sim}{\mathbf{C}}$ is a 4×4 diagonal matrix with $C_{11} = C_c$, $C_{22} = 0$, and $C_{33} = C_{44} = C_s$.

$$C_\sigma = \frac{-2}{1 + 2i\beta_\sigma \Sigma_{m=1}^{M} \alpha_m Z_{\sigma m}} \tag{A-6}$$

and

$$Z_{\sigma m} = \frac{\sin(\omega\Delta t) + i\beta_\sigma \Delta x |k_\perp| \left[1 - \gamma_m \sin^2\left(\dfrac{\omega\Delta t}{2}\right)\right]}{4\sin^2\left(\dfrac{\omega\Delta t}{2}\right) + i\gamma_m \beta_\sigma \Delta x |k_\perp| \sin(\omega\Delta t) - (\beta_\sigma \Delta x k_\parallel)^2}. \tag{A-7}$$

In equations (A-6) and (A-7), σ stands for either c or s.

The matrices $\underset{\sim}{\mathbf{M}}_\pm$ which give the transformation from the incident (+) and reflected (−) Cartesian components of displacement $\tilde{\mathbf{u}}$ to the normal modes of the interior mesh finite-difference equations $\tilde{\mathbf{U}}$ are defined through equations (35) and (36). The displacement fields satisfy some finite-difference analog of equation (11). Several finite-difference schemes are available. The scheme chosen here is the familiar second-order displacement formulation in which accurate, centered second-order finite-difference operators replace all derivatives. It not necessary to reproduce the finite-difference equations explicitly because of their familiarity. If solutions of the form

$$\mathbf{u} = \tilde{\mathbf{u}} \exp\left[i(k_x x + k_y y + k_z z - \omega t) \right] \tag{A-8}$$

are assumed, then the finite-difference analog of equation (11) reduces to

$$\underset{\sim}{\mathbf{W}}\tilde{\mathbf{u}} = 0, \tag{A-9}$$

where $\underset{\sim}{\mathbf{W}}$ is a symmetric 3×3 matrix. A typical diagonal component is

$$W_{xx} = 4 \sin^2\left(\frac{\omega \Delta t}{2}\right) - 4\beta_c^2 \sin^2\left(\frac{k_x \Delta x}{2}\right) - 4\beta_s^2\left[\sin^2\left(\frac{k_y \Delta y}{2}\right) + \sin^2\left(\frac{k_z \Delta z}{2}\right)\right], \tag{A-10}$$

with analogous expressions for W_{yy} and W_{zz}. A typical off-diagonal component is

$$W_{xy} = W_{yx} = -(\beta_c^2 - \beta_s^2) \sin(k_x \Delta x) \sin(k_y \Delta y). \tag{A-11}$$

Solution of the dispersion relation

$$\det(\underset{\sim}{\mathbf{W}}) = 0 \tag{A-12}$$

yields three pairs of solutions for k_x, each pair corresponding to rightgoing and leftgoing (+) and (−) components of one of three normal modes $\tilde{\mathbf{u}}_c$, $\tilde{\mathbf{u}}_1$, and $\tilde{\mathbf{u}}_2$ of the interior mesh finite-difference equations. The Cartesian components of the normal mode are determined by equation (A-9) to within an arbitrary normalization constant. These normalization constants affect the magnitude of the off-diagonal elements of the reflection matrix $\underset{\sim}{\mathbf{R}}$, i.e., those elements which represent conversion from one normal mode into another. To be consistent with Clayton and Engquist (1977), the magnitude of the compressional mode is normalized to k_c and that of the shear modes to k_s. From equation (44), it follows that the transformation matrices $\underset{\sim}{\mathbf{M}}_{\pm}$ may be formed from the column vectors representing the components of the three normal modes:

$$\underset{\sim}{\mathbf{M}}_{\pm} = \{\tilde{\mathbf{u}}_{c\pm}, \tilde{\mathbf{u}}_{1\pm}, \tilde{\mathbf{u}}_{2\pm}\}^{-1}. \tag{A-13}$$

For simplicity, equal grid spacings are chosen along the three axes, $\Delta y = \Delta z = \Delta x$, for the numerical results shown in the text.

APPENDIX B

PARAXIAL APPROXIMATION IN THREE DIMENSIONS

The paraxial approximation absorbing boundary condition for the elastic wave equation in three dimensions apparently has not appeared elsewhere in the literature. Since the generalization to three dimensions from two is not immediately obvious, it is given here. The exact radiation condition for 3-D elastic waves is first derived in the frequency-wavenumber domain. This radiation condition is expanded as a power series in transverse (to the boundary) wavenumber, retaining terms up to second order. Application of the transformations $-i\omega \leftrightarrow \partial/\partial t$, $i\mathbf{k} \leftrightarrow \nabla$ to the approximate radiation condition yields the second-order paraxial approximation absorbing boundary for elastic waves in three dimensions.

Consider an isotropic, homogeneous elastic medium with compressional and shear velocities c and s, respectively. The three components of displacement $\mathbf{u} = (u_x, u_y, u_z)$ in Cartesian coordinates may be expressed in terms of three potentials ϕ, ψ, and χ, each of which satisfies a scalar wave equation (Morse and Feshbach, 1953). Let all field quantities vary as $\exp[i(k_y y + k_z z - \omega t)]$. Then

$$\mathbf{u}(x, y, z, t) = \tilde{\mathbf{u}}(x) \exp[i(k_y y + k_z z - \omega t)] \tag{B-1}$$

$$\mathbf{\Phi}(x, y, z, t) = \tilde{\mathbf{\Phi}}(x) \exp[i(k_y y + k_z z - \omega t)], \tag{B-2}$$

where $\mathbf{\Phi}$ is a vector whose components are the three scalar potentials $\mathbf{\Phi} = (\phi, \psi, \chi)$. The x variation of $\tilde{\mathbf{\Phi}}$ is given by

$$\tilde{\mathbf{\Phi}}(x) = \tilde{\mathbf{\Phi}}(0) \exp(i\underset{\sim}{\mathbf{K}}x), \tag{B-3}$$

where $\underset{\sim}{\mathbf{K}}$ is the diagonal matrix

$$\underset{\sim}{\mathbf{K}} = \begin{bmatrix} k_{xc} & 0 & 0 \\ 0 & k_{xs} & 0 \\ 0 & 0 & k_{xs} \end{bmatrix}, \tag{B-4}$$

and $k_{xc}^2 = (k_c^2 - k_\perp^2)$, $k_{xs}^2 = (k_s^2 - k_\perp^2)$, $k_c = \omega/c$, $k_s = \omega/s$, and $k_\perp^2 = (k_y^2 + k_z^2)$. The positive roots are chosen for k_{xc} and k_{xs} for propagation in the positive x direction. The displacement vector $\tilde{\mathbf{u}}$ and the potential vector $\tilde{\mathbf{\Phi}}$ are related by the transformations

$$\tilde{\mathbf{u}} = \underset{\sim}{\mathbf{D}}\tilde{\mathbf{\Phi}} \qquad \tilde{\mathbf{\Phi}} = \underset{\sim}{\mathbf{D}}^{-1}\tilde{\mathbf{u}}, \tag{B-5}$$

where

$$\underset{\sim}{\mathbf{D}} = \begin{bmatrix} k_{xc} & 0 & (k_y^2 + k_z^2)/k_s \\ k_y & -k_z & -(k_y k_{xs})/k_s \\ k_z & k_y & -(k_z k_{xs})/k_s \end{bmatrix} \tag{B-6}$$

and

$$\underset{\sim}{D}^{-1} = (k_{xc}k_{xs} + k_\perp^2)^{-1}$$

$$\times \begin{bmatrix} k_{xs} & k_y & k_z \\ 0 & -k_z\left(1 + \frac{k_{xc}k_{xs}}{k_\perp^2}\right) & k_y\left(1 + \frac{k_{xc}k_{xs}}{k_\perp^2}\right) \\ k_s & -\frac{k_y k_{xc} k_s}{k_\perp^2} & -\frac{k_z k_{xc} k_s}{k_\perp^2} \end{bmatrix}. \quad \text{(B-7)}$$

The displacement vector may be differentiated as

$$\frac{\partial \tilde{\mathbf{u}}}{\partial x} = \underset{\sim}{D}\frac{\partial \tilde{\mathbf{\Phi}}}{\partial x} = i\underset{\sim}{D}\underset{\sim}{K}\tilde{\mathbf{\Phi}} = i\underset{\sim}{D}\underset{\sim}{K}\underset{\sim}{D}^{-1}\tilde{\mathbf{u}} = i\underset{\sim}{Q}\tilde{\mathbf{u}}, \quad \text{(B-8)}$$

where

$$\underset{\sim}{Q} = \underset{\sim}{D}\underset{\sim}{K}\underset{\sim}{D}^{-1} = (k_{xc}k_{xs} + k_\perp^2)^{-1} \begin{bmatrix} k_{xs}k_c^2 & k_y k_{xc}(k_{xc} - k_{xs}) & k_z k_{xc}(k_{xc} - k_{xs}) \\ k_y k_{xs}(k_{xc} - k_{xs}) & k_y^2 k_{xc} + k_z^2 k_{xs} + k_{xs}^2 k_{xc} & k_y k_z (k_{xc} - k_{xs}) \\ k_z k_{xs}(k_{xc} - k_{xs}) & k_y k_z (k_{xc} - k_{xs}) & k_y^2 k_{xs} + k_z^2 k_{xc} + k_{xs}^2 k_{xc} \end{bmatrix}. \quad \text{(B-9)}$$

The second-order paraxial approximation is now made. Only terms of up to second order in k_y and k_z (assumed small) are retained in equation (B-9). For example, k_{xc} and k_{xs} are expanded as

$$k_{xc} \approx k_c - \frac{k_\perp^2}{2k_c} \quad \text{(B-10)}$$

and

$$k_{xs} \approx k_s - \frac{k_\perp^2}{2k_s}. \quad \text{(B-11)}$$

The component form of equation (B-8) becomes

$$-\left(\tilde{u}_x + \frac{i}{k_c}\frac{\partial \tilde{u}_x}{\partial x}\right) = \frac{k_\perp^2 \tilde{u}_x}{2k_c}\left(\frac{1}{k_c} - \frac{2}{k_s}\right) + \left(\frac{1}{k_s} - \frac{1}{k_c}\right)(k_y\tilde{u}_y + k_z\tilde{u}_z), \quad \text{(B-12)}$$

$$-\left(\tilde{u}_y + \frac{i}{k_s}\frac{\partial \tilde{u}_y}{\partial x}\right) = \left(k_y\tilde{u}_x + \frac{k_y k_z \tilde{u}_z}{k_s}\right)\left(\frac{1}{k_s} - \frac{1}{k_c}\right) + \frac{k_y^2\tilde{u}_y}{2k_s}\left(\frac{1}{k_s} - \frac{2}{k_c}\right) - \frac{k_z^2\tilde{u}_y}{2k_s^2}, \quad \text{(B-13)}$$

and

$$-\left(\tilde{u}_z + \frac{i}{k_s}\frac{\partial \tilde{u}_z}{\partial x}\right) = \left(k_z\tilde{u}_x + \frac{k_y k_z \tilde{u}_y}{k_s}\right)\left(\frac{1}{k_s} - \frac{1}{k_c}\right) + \frac{k_z^2\tilde{u}_z}{2k_s}\left(\frac{1}{k_s} - \frac{2}{k_c}\right) - \frac{k_y^2\tilde{u}_z}{2k_s^2}. \quad \text{(B-14)}$$

The 3-D paraxial absorbing boundary condition for the elastic wave equation is obtained by multiplying through by ω^2 in equations (B-12)–(B-14) and transforming back into the time-space domain:

$$\frac{\partial^2 u_x}{\partial t^2} + c\frac{\partial^2 u_x}{\partial x \partial t} = \left(cs - \frac{c^2}{2}\right)\left(\frac{\partial^2 u_x}{\partial y^2} + \frac{\partial^2 u_x}{\partial z^2}\right) - (c - s)\left(\frac{\partial^2 u_y}{\partial t \partial y} + \frac{\partial^2 u_z}{\partial t \partial z}\right); \quad \text{(B-15)}$$

$$\frac{\partial^2 u_y}{\partial t^2} + s\frac{\partial^2 u_y}{\partial x \partial t} = \left(cs - \frac{s^2}{2}\right)\frac{\partial^2 u_y}{\partial y^2} + \frac{s^2}{2}\frac{\partial^2 u_y}{\partial z^2} - (c - s)\frac{\partial^2 u_x}{\partial t \partial y} + (cs - s^2)\frac{\partial^2 u_z}{\partial y \partial z}; \quad \text{(B-16)}$$

and

$$\frac{\partial^2 u_z}{\partial t^2} + s\frac{\partial^2 u_z}{\partial x \partial t} = \left(cs - \frac{s^2}{2}\right)\frac{\partial^2 u_z}{\partial z^2} + \frac{s^2}{2}\frac{\partial^2 u_z}{\partial y^2} - (c - s)\frac{\partial^2 u_x}{\partial t \partial z} + (cs - s^2)\frac{\partial^2 u_y}{\partial y \partial z}. \quad \text{(B-17)}$$